12TH CHEMISTRY · CHAPTER 17 · FULL CHAPTER NOTES

# Group 2 Elements

Topic-wise short and long questions with answers from the complete chapter

## INTRODUCTION

### Q.1 [Short] What are Group 2 elements? Why are they called alkaline earth metals?

Group 2 elements are s-block elements of the periodic table: beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr) and barium (Ba). They are also called the **alkaline earth metals** because of their metallic nature and occurrence in the earth's crust. Their reactivity lies between the highly reactive alkali metals and the less reactive transition metals. All have two electrons in their outermost s-subshell ($ns^2$), so they show similar chemical properties.

### Q.2 [Short] Write some general uses of the Group 2 elements given at the start of the chapter.

Ca and Mg are among the eight most widely occurring elements in the earth's crust. Be is used to make copper alloys harder for aerospace materials. Mg is an important part of chlorophyll, the green matter in plants. Mg, Ca, Sr and Ba are essential components of fireworks due to their vibrant flame colours.

## 17.1 PROPERTIES AND TRENDS OF GROUP 2 ELEMENTS

### Q.3 [Short] Why do Group 2 elements show similar chemical properties?

All of them have two electrons in their outermost s-subshell ($ns^2$). The same valence-shell configuration gives them similar chemical properties, although the properties change gradually from top to bottom.

### Q.4 [Long] Give the salient properties of Group 2 elements (Be–Ba) in tabular form and state the main trends.

| Property | Be | Mg | Ca | Sr | Ba |
|---|---|---|---|---|---|
| Atomic number | 4 | 12 | 20 | 38 | 56 |
| Electron configuration | [He] $2s^2$ | [Ne] $3s^2$ | [Ar] $4s^2$ | [Kr] $5s^2$ | [Xe] $6s^2$ |
| Metallic radius (pm) | 112 | 160 | 197 | 215 | 222 |
| Ionic radius $M^{2+}$ (pm) | 31 | 65 | 99 | 113 | 135 |
| 1st ionization energy | 900 | 738 | 590 | 550 | 503 |
| 2nd ionization energy | 1757 | 1451 | 1145 | 1064 | 965 |
| Electronegativity | 1.5 | 1.2 | 1.0 | 1.0 | 0.9 |
| Melting point (°C) | 1287 | 650 | 842 | 777 | 727 |

| Property | Be | Mg | Ca | Sr | Ba |
|---|---|---|---|---|---|
| Density (g/cm$^3$) | 1.85 | 1.74 | 1.54 | 2.64 | 3.51 |
| Standard electrode potential (E°) | –1.97 | –2.34 | –2.87 | –2.89 | –2.91 |

- Metallic and ionic radii **increase** down the group (a new shell is added each time).
- First and second ionization energies and electronegativity **decrease** down the group.
- E° becomes more negative down the group, so the metals become stronger reducing agents.
- Melting point and density do not change regularly; this is related to the different packing of the metal atoms down the group.

**Q.5** [Short] **Why does the ionization energy of Group 2 elements decrease down the group?**

Down the group the atomic size increases and the shielding effect of the inner electrons increases. The outer electrons are held less strongly by the nucleus, so less energy is needed to remove them (900 for Be down to 503 for Ba).

**Q.6** [Short] **Why is the second ionization energy of every Group 2 element higher than the first?**

After the first electron is removed, the ion $M^+$ has a positive charge and the same nuclear charge holds fewer electrons more strongly. So more energy is needed to remove the second electron (e.g., Mg: 738 and 1451).

**Q.7** [Short] **Why is the trend of melting point and density in Group 2 irregular?**

According to the textbook, the trend in melting point and density is related to the different packing of the metal atoms down the group. So the values do not fall or rise smoothly: Mg has the lowest melting point (650 °C) and Ca the lowest density (1.54 g/cm$^3$).

## 17.1.1 Chemical reactivity of Group 2 Elements

**Q.8** [Short] **Explain the reactivity of Group 2 elements in terms of their electron configuration. Give the reactivity series.**

Group 2 atoms have two electrons in their outermost subshell ($ns^2$). After losing these two electrons they become stable ions with a +2 charge ($M^{2+}$), so they react quickly with other substances. Reactivity **increases from Be to Ba**, because down the group the atomic size and shielding effect increase and the ionization energy decreases, so the removal of electrons becomes easier.

Reactivity series: Be < Mg < Ca < Sr < Ba

**Q.9** [Short] **Which is the least reactive and which is the most reactive element of Group 2 (Be–Ba)? Why?**

Be is the least reactive and Ba the most reactive. Be is the smallest atom with the highest ionization energy, so it loses electrons with difficulty. Ba is the largest atom with the most shielding and the lowest ionization energy, so it loses its electrons most easily.

## 17.2 CHEMICAL REACTIONS OF GROUP 2 ELEMENTS

### 17.2.1 Reactions with Oxygen

**Q.10** [Long] **Describe the reactions of Group 2 elements with oxygen. What is the nature of the oxides formed?**

All Group 2 metals burn in oxygen to form metal oxides (MO).

$$2M + O_2 \rightarrow 2MO$$

- **Beryllium** reacts with oxygen only at high temperature (above 600 °C) to form BeO. A thin, strong oxide layer usually protects it, so it is less reactive than the other members: $2Be_{(s)} + O_{2(g)} \rightarrow 2BeO_{(s)}$
- **Magnesium** burns brightly in air or oxygen to form MgO: $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$
- **Calcium** reacts more vigorously, with a brick-red flame: $2Ca_{(s)} + O_{2(g)} \rightarrow 2CaO_{(s)}$

The oxides are generally white solids and are **basic**; their basic character increases down the group. The basic character in water is due to the reaction of the oxide ion with water to produce $OH^-$ ions:

$$O^{2-}_{(s)} + H_2O_{(l)} \rightarrow 2OH^-_{(aq)}$$

**Q.11** [Short] **Why is beryllium less reactive towards oxygen than the other members of the group?**

A thin, strong oxide layer usually protects beryllium, so it reacts with oxygen only at high temperature (above 600 °C).

**Q.12** [Short] **What is a basic oxide? Why are Group 2 oxides basic in water?**

A basic oxide is a metal oxide that reacts with water to give an alkaline solution (and with acids to give salt and water). Group 2 oxides are basic because the oxide ion reacts with water to produce $OH^-$ ions: $O^{2-} + H_2O \rightarrow 2OH^-$. The basic character increases down the group.

**Q.13** [Short] **What colours do Group 2 elements give in a flame?**

Magnesium burns with a bright white light, calcium gives a brick-red flame, strontium a red (crimson) flame and barium a green flame (Figure 17.2). This is why they are used in fireworks.

## 17.2.2 Reactions with Water

### Q.14 [Long] Describe the reactions of Group 2 elements with water.

Except beryllium, Group 2 metals react with water to produce metal hydroxides, $M(OH)_2$, and hydrogen gas.

- **Beryllium** does not react with water or steam, even when heated, because of its protective oxide layer.
- **Magnesium** reacts very slowly with cold water, forming $Mg(OH)_2$ and $H_2$. With steam it reacts faster, forming MgO and hydrogen.
- **Calcium** reacts with cold water and steam more vigorously than magnesium, forming $Ca(OH)_2$ and $H_2$. Calcium hydroxide is sparingly soluble and forms a white precipitate.

$$\text{Cold water: } Mg_{(s)} + 2H_2O_{(l)} \rightarrow Mg(OH)_{2(aq)} + H_{2(g)}$$
$$\text{Steam: } Mg_{(s)} + H_2O_{(g)} \rightarrow MgO_{(s)} + H_{2(g)}$$
$$Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + H_{2(g)}$$

The hydroxides of Group 2 are less basic than those of Group 1, but their basic character increases down the group due to increasing solubility.

### Q.15 [Short] Magnesium gives different products with cold water and with steam. Explain with equations.

With cold water magnesium reacts very slowly to form the hydroxide. With steam it reacts faster and forms the oxide.

$$Mg_{(s)} + 2H_2O_{(l)} \rightarrow Mg(OH)_{2(aq)} + H_{2(g)}$$
$$Mg_{(s)} + H_2O_{(g)} \rightarrow MgO_{(s)} + H_{2(g)}$$

### Q.16 [Short] Why does a white precipitate appear when calcium reacts with water?

The product, calcium hydroxide $Ca(OH)_2$, is only sparingly soluble in water, so it appears as a white precipitate.

## 17.2.3 Reactions with Acids

### Q.17 [Long] Describe the reactions of Group 2 metals with dilute acids.

Group 2 metals react with dilute acids to produce a salt and hydrogen gas. Reactivity increases down the group.

$$M_{(s)} + 2HCl_{(aq)} \rightarrow MCl_{2(aq)} + H_{2(g)}$$

- **Beryllium** forms $BeCl_2$ and hydrogen, but its protective oxide layer can slow the reaction.
- **Magnesium** reacts readily: $Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$; $Mg_{(s)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + H_{2(g)}$
- **Calcium** reacts vigorously: $Ca_{(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_{2(g)}$

- With sulphuric acid, the reaction of calcium can stop early, because the $CaSO_4$ formed is sparingly soluble and coats the metal with a protective layer: $Ca_{(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + H_{2(g)}$

**Q.18** [Short] **Why can the reaction of calcium with sulphuric acid stop early?**

The calcium sulphate ($CaSO_4$) formed is sparingly soluble. It coats the metal surface with a protective layer, which keeps the acid away from the metal.

## Quick Check 17.1

**Q.19** [Short] **a) Write equations for the following reactions: i) Ca with oxygen ii) Ba with water**

$$\text{i) } 2Ca_{(s)} + O_{2(g)} \rightarrow 2CaO_{(s)}$$
$$\text{ii) } Ba_{(s)} + 2H_2O_{(l)} \rightarrow Ba(OH)_{2(aq)} + H_{2(g)}$$

**Q.20** [Short] **b) Predict the extent of the reactions of Sr and Ba with $H_2SO_4$.**

They take place only to a very small extent. $SrSO_4$ and $BaSO_4$ are even less soluble than $CaSO_4$, so an insoluble sulphate layer coats the metal almost at once and stops the reaction.

**Q.21** [Short] **c) Explain why Group 2 elements are reactive, based on their electron configuration and ion formation.**

They have two electrons in the outermost s-subshell ($ns^2$). By losing them they form stable $M^{2+}$ ions, so they react quickly with other substances.

**Q.22** [Short] **d) Write a balanced equation for calcium and barium burning in air and compare their reactivity.**

$$2Ca_{(s)} + O_{2(g)} \rightarrow 2CaO_{(s)}$$
$$2Ba_{(s)} + O_{2(g)} \rightarrow 2BaO_{(s)}$$

Barium is more reactive: it is larger, has more shielding and a lower ionization energy, so it loses electrons more easily.

**Q.23** [Short] **e) Predict the density and melting point of the radioactive radium (Ra) following the trend in Group 2.**

From Ca to Ba the density rises (1.54, 2.64, 3.51 $g/cm^3$) and the melting point falls (842, 777, 727 °C). So radium should have a density above 3.51 $g/cm^3$ (about 5 $g/cm^3$) and a melting point a little below 727 °C (about 700 °C).

## 17.3 COMPARISON OF GROUP 2 ELEMENTS WITH GROUP 1 ELEMENTS

### Q.24 [Long] Compare the properties and reactivity of Group 2 elements with Group 1 elements.

Both groups contain highly reactive metals, but they differ because of their ionization energies. Both show increasing reactivity down the group due to increasing atomic size and decreasing ionization energy.

| Feature | Group 1 (Alkali Metals) | Group 2 (Alkaline Earth Metals) |
|---|---|---|
| Valence electrons | 1 ($ns^1$) | 2 ($ns^2$) |
| Ion formation | Forms +1 ions ($M^+$) | Forms +2 ions ($M^{2+}$) |
| Reactivity | More reactive | Less reactive (compared to Group 1 elements in the same period) |
| Reason for reactivity | Low ionization energies | Higher ionization energies |
| Hardness | Very soft (can be cut with a knife) | Harder than Group 1 metals |
| Density | Low densities | Higher densities than Group 1 metals |
| Melting/boiling points | Lower | Higher |
| Reaction with water | Very vigorous, often explosive (e.g., Na with water) | Less vigorous (e.g., Ca with water, Mg with steam) |
| Oxides | Form $M_2O$ (e.g., $Na_2O$); highly basic | Form MO (e.g., CaO); less basic than Group 1 oxides |
| Hydroxides | Highly soluble, strong bases | Less soluble, but solubility increases down the group; less basic |

### Q.25 [Short] Why are Group 1 metals more reactive than Group 2 metals of the same period?

Group 1 metals have only one electron to lose, which makes them extremely eager to react and form a +1 ion. Group 2 metals lose two electrons, which requires more energy, so they are reactive but generally less so.

### Q.26 [Short] Why are Group 2 metals harder, denser and higher melting than Group 1 metals?

Each Group 2 atom gives two electrons to the metallic bond and its ion is smaller with a +2 charge, so the metallic bonding is stronger and the atoms are packed more closely. This makes the metals harder and denser with higher melting points.

*The textbook gives these differences in Table 17.2 without the reason; the reason given here is standard chemistry.*

## Quick Check 17.2 (parts a, b)

**Q.27** [Short] **a) State two physical differences between Group 1 and Group 2 metals.**

(i) Group 2 metals are harder; Group 1 metals are very soft and can be cut with a knife.
(ii) Group 2 metals have higher densities and higher melting/boiling points.

**Q.28** [Short] **b) State the general formula for a Group 1 oxide and a Group 2 oxide.**

Group 1 oxide: $M_2O$ (e.g., $Na_2O$). Group 2 oxide: MO (e.g., CaO).

## 17.4 THERMAL DECOMPOSITION OF CARBONATES AND NITRATES

**Q.29** [Short] **What is thermal decomposition?**

**Thermal decomposition** is a chemical reaction in which a single compound breaks down into two or more simpler substances when heated. It usually requires energy (endothermic) to break the chemical bonds within the compound.

**Q.30** [Long] **Explain the thermal decomposition of Group 2 carbonates and the trend in their thermal stability.**

Group 2 carbonates decompose on heating to form a metal oxide (MO) and carbon dioxide. Their thermal stability **increases down the group**: $BeCO_3$ decomposes at a lower temperature than $MgCO_3$, which decomposes at a lower temperature than $CaCO_3$, and so on. $BaCO_3$ requires the highest temperature.

$$MgCO_{3(s)} \rightarrow MgO_{(s)} + CO_{2(g)}$$
$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$$

**Reason:** The trend is due to the size of the $M^{2+}$ ion. Smaller, highly charged cations (like $Be^{2+}$) have a greater **polarizing power**. They distort the large carbonate ion ($CO_3^{2-}$) to a larger extent, weakening its bonds and making it easier to decompose. As the metal ion gets larger down the group, its polarizing power decreases, so the carbonates become more stable and need more heat.

| Metal carbonate | $BeCO_3$ | $MgCO_3$ | $CaCO_3$ | $SrCO_3$ | $BaCO_3$ |
|---|---|---|---|---|---|
| Decomposition temperature (°C) at 1 atm | 250 | 540 | 900 | 1289 | 1360 |

**Q.31** [Long] **Explain the thermal decomposition of Group 2 nitrates and the trend in their thermal stability.**

Group 2 nitrates decompose on heating to form a metal oxide (MO), brown nitrogen dioxide gas ($NO_2$) and oxygen ($O_2$).

$$2Ca(NO_3)_{2(s)} \rightarrow 2CaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

Like the carbonates, the thermal stability of the nitrates **increases down the group**: $Be(NO_3)_2$ is the least stable and $Ba(NO_3)_2$ the most stable. The same polarizing-power argument applies: the smaller $M^{2+}$ ions at the top of the group distort the nitrate ion ($NO_3^-$) more effectively, so decomposition is easier at lower temperatures.

### Q.32 [Short] What is meant by polarizing power? On what does it depend?

Polarizing power is the ability of a cation to distort the electron cloud of a large anion. It is greater for smaller and more highly charged cations (high charge density). In Group 2 it decreases from $Be^{2+}$ to $Ba^{2+}$, because the ions get larger while the charge stays +2.

### Q.33 [Short] How is thermal decomposition useful in the analysis of Group 2 compounds?

By observing the temperature at which a Group 2 carbonate or nitrate decomposes, chemists can identify an unknown compound and confirm the presence of a specific element. The decomposition pattern also helps to understand the relationship between ionic size, charge density and thermal stability.

### Q.34 [Short] What would you observe when a Group 2 nitrate is heated?

A white solid residue of the metal oxide is left, brown fumes of nitrogen dioxide ($NO_2$) are given off, and oxygen is released (it relights a glowing splint).

## Quick Check 17.2 (parts c, d, e)

### Q.35 [Short] c) Write the balanced equation for the thermal decomposition of $SrCO_3$.

$$SrCO_{3(s)} \rightarrow SrO_{(s)} + CO_{2(g)}$$

### Q.36 [Short] d) Explain qualitatively why magnesium carbonate is thermally less stable than calcium carbonate.

$Mg^{2+}$ is smaller than $Ca^{2+}$, so it has greater polarizing power. It distorts the carbonate ion more and weakens its bonds, so $MgCO_3$ decomposes at 540 °C while $CaCO_3$ needs 900 °C.

### Q.37 [Short] e) Write the equations for the thermal decomposition of i) magnesium nitrate ii) Barium nitrate. Which of these decomposes at a lower temperature?

$$\text{i) } 2Mg(NO_3)_{2(s)} \rightarrow 2MgO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$
$$\text{ii) } 2Ba(NO_3)_{2(s)} \rightarrow 2BaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

Magnesium nitrate decomposes at the lower temperature, because the smaller $Mg^{2+}$ ion distorts the nitrate ion more.

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## 17.5 SOLUBILITY OF GROUP 2 SULPHATES AND HYDROXIDES

### Q.38 [Long] Define enthalpy of hydration, lattice enthalpy and enthalpy of solution. How do they decide solubility?

The solubility of an ionic compound in water depends on the balance between two energy changes:

1. **Enthalpy of hydration ($\Delta H_{hyd}$):** the energy released when gaseous ions are surrounded and attracted by water molecules. It is always exothermic. Smaller ions with higher charge density attract water more strongly, so their $\Delta H_{hyd}$ is more negative. It becomes less negative down a group as the ions get larger.
2. **Lattice enthalpy ($\Delta H_{lattice}$):** the energy required to break apart one mole of a solid ionic compound into its gaseous ions. It is always endothermic. It depends on the charges and sizes of the ions: when cation and anion are comparable in size the forces are stronger and the lattice energy is higher; when one ion is much larger than the other the forces are weaker.

The **enthalpy of solution ($\Delta H_{soln}$)** is the overall energy change when a solid ionic substance dissolves. A more negative or less positive $\Delta H_{soln}$ generally favours higher solubility.

### Q.39 [Long] Explain the trends in the solubility of Group 2 hydroxides and sulphates down the group.

**Hydroxides:** solubility **increases down the group** ($Mg(OH)_2$ is sparingly soluble; $Ba(OH)_2$ is quite soluble). Down the group both $\Delta H_{hyd}$ (of $M^{2+}$) and $\Delta H_{lattice}$ decrease, but the decrease in lattice enthalpy is more significant than the decrease in hydration enthalpy. So less energy is needed to break the lattice, and dissolving becomes more favourable.

**Sulphates:** solubility **decreases down the group** ($MgSO_4$ is soluble; $BaSO_4$ is insoluble). The sulphate ion ($SO_4^{2-}$) is relatively large. Down the group both enthalpies decrease, but for a large anion the decrease in hydration enthalpy is more significant than the decrease in lattice enthalpy. So the energy released on hydration falls faster, and dissolving becomes less favourable.

| Hydroxide | Solubility at 298 K (mol/100 g water) | Sulphate | Solubility at 298 K (mol/100 g water) |
|---|---|---|---|
| $Mg(OH)_2$ | $2 \times 10^{-5}$ | $MgSO_4$ | $1.83 \times 10^{-1}$ |
| $Ca(OH)_2$ | $1.5 \times 10^{-3}$ | $CaSO_4$ | $4.66 \times 10^{-3}$ |
| $Sr(OH)_2$ | $3.4 \times 10^{-3}$ | $SrSO_4$ | $7.11 \times 10^{-5}$ |
| $Ba(OH)_2$ | $1.5 \times 10^{-2}$ | $BaSO_4$ | $9.43 \times 10^{-7}$ |

**Q.40** [Short] **Why does hydration enthalpy become less negative down Group 2?**

Down the group the ions get larger, so their charge density falls. They attract water molecules less strongly and less energy is released on hydration.

**Q.41** [Short] **How does the relative size of cation and anion affect lattice energy?**

When the cation and anion are comparable in size, the attractive forces are stronger and the lattice energy is higher. When one ion is much larger than the other, the forces are weaker and the lattice energy is lower. So with smaller Group 2 ions the lattice energy of the hydroxides is higher, and vice versa.

**Q.42** [Short] **The solubility trends of Group 2 hydroxides and sulphates are opposite. Why?**

Both enthalpies decrease down the group, but which one decreases more depends on the anion. With the small hydroxide ion the lattice enthalpy falls more, so solubility increases. With the large sulphate ion the hydration enthalpy falls more, so solubility decreases.

**Q.43** [Short] **Why does the basic character of Group 2 hydroxides increase down the group?**

Their solubility increases down the group, so more $OH^-$ ions are released into the solution and the solution is more alkaline.

## Quick Check 17.3

**Q.44** [Short] **a) Why does $MgSO_4$ dissolve easily as compared to $BaSO_4$?**

$Mg^{2+}$ is small, so its hydration enthalpy is highly negative and compensates the lattice enthalpy. $Ba^{2+}$ is large, so its hydration enthalpy is much less negative while the lattice enthalpy has fallen only a little; dissolving $BaSO_4$ is unfavourable.

**Q.45** [Short] **b) How does lattice energy affect the solubility of Group 2 sulphates?**

Higher lattice energy means lower solubility. In the sulphates the lattice energy decreases only slightly down the group, while the hydration enthalpy decreases much more, so the lattice energy is less and less compensated and solubility decreases.

**Q.46** [Short] **c) Which hydroxide is more soluble: $Ca(OH)_2$ or $Sr(OH)_2$? Give reason.**

$Sr(OH)_2$ ($3.4 \times 10^{-3}$ against $1.5 \times 10^{-3}$ mol/100 g of water). Hydroxide solubility increases down the group because lattice enthalpy decreases more than hydration enthalpy.

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## 17.6 COMPLEX ION FORMATION

### Q.47 [Short] What is a complex ion? Define coordination number.

A **complex ion** is an ion in which a central metal is bonded to a number of surrounding electron-donating species. The central metal ion acts as a Lewis acid (electron pair acceptor); the molecules or ions with lone pairs (e.g., $H_2O$, $NH_3$, $Cl^-$, $OH^-$) act as Lewis bases. The number of coordinate bonds between the central metal ion and the electron donors is the **coordination number**; common values are 2, 4 and 6.

### Q.48 [Long] Discuss the formation of complex ions by Group 2 elements.

- Complex formation is more typical of transition metals (available d-orbitals), but Group 2 elements, especially those at the top of the group, also form complex ions.
- The ability depends on size and charge. Smaller ions have a higher charge density, so they attract electron-donating species better. All Group 2 ions have a +2 charge.
- In water the ions become hydrated; these are complex ions with water as the electron pair donor. $Be^{2+}$, the smallest ion, forms $[Be(H_2O)_4]^{2+}$ (coordination number 4). $Mg^{2+}$ is larger and usually forms $[Mg(H_2O)_6]^{2+}$ (coordination number 6).
- $Ca^{2+}$, $Sr^{2+}$ and $Ba^{2+}$ also form hydrated ions, mostly with coordination number 6, but the strength of the interaction decreases down the group.
- With donors having more than one donatable electron pair, ring-like structures form. Calcium forms a complex with the six-pair donor EDTA: $Ca^{2+}_{(aq)} + EDTA^{4-}_{(aq)} \rightarrow [Ca(EDTA)]^{2-}_{(aq)}$
- **Use:** EDTA removes $Ca^{2+}$ and $Mg^{2+}$ ions from hard water (water softening) and can remove toxic metal ions from the body.

### Q.49 [Short] Why does $Be^{2+}$ form complexes more easily than $Ba^{2+}$? Why is its coordination number only 4?

$Be^{2+}$ is the smallest Group 2 ion, so it has the highest charge density and attracts electron pair donors most strongly. Being so small, it has room for only four water molecules, $[Be(H_2O)_4]^{2+}$; the larger $Mg^{2+}$ takes six, $[Mg(H_2O)_6]^{2+}$.

### Q.50 [Short] Why do Group 2 elements form fewer complexes than transition metals?

Transition metals have available d-orbitals for bonding with electron pair donors. Group 2 ions do not, so their tendency to form complexes is far less, though still significant in some cases.

### Q.51 [Short] What is EDTA? Give its uses.

EDTA (ethylenediaminetetraacetic acid) is a six electron pair donor that forms a ring-structured complex with calcium, $[Ca(EDTA)]^{2-}$. It is used in water softening to remove $Ca^{2+}$ and $Mg^{2+}$ ions from hard water, and in medicines to remove toxic metal ions from the body.

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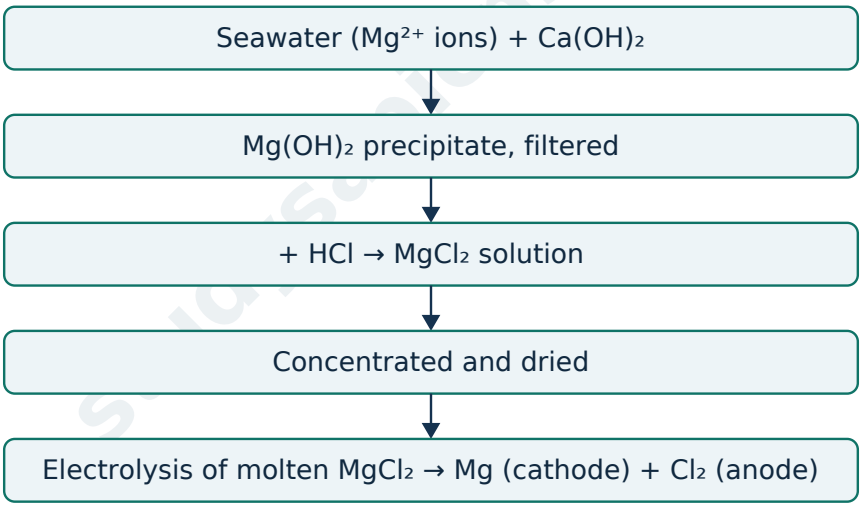
## 17.7 EXTRACTION AND PURIFICATION

### Q.52 [Long] Describe the production of magnesium from seawater by the Dow process.

The **Dow process** is a multi-step chemical and electrochemical process that extracts magnesium metal from the vast reserves of $Mg^{2+}$ ions in seawater.

1. **Precipitation:** Seawater is treated with calcium hydroxide, $Ca(OH)_2$ (from limestone or oyster shells), which precipitates magnesium as insoluble magnesium hydroxide.
2. **Conversion to chloride:** The $Mg(OH)_2$ is filtered and reacted with hydrochloric acid to form magnesium chloride.
3. **Concentration and drying:** The $MgCl_2$ solution is concentrated and dried.
4. **Electrolysis:** Molten $MgCl_2$ is electrolysed in large cells. Pure magnesium metal is produced at the cathode and chlorine gas at the anode.

$$Mg^{2+}_{(aq)} + Ca(OH)_{2(s)} \rightarrow Mg(OH)_{2(s)} + Ca^{2+}_{(aq)}$$
$$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$
$$Mg^{2+}_{(l)} + 2Cl^{-}_{(l)} \rightarrow Mg_{(l)} + Cl_{2(g)} \text{ (electrolysis)}$$




*Dow process*

### Q.53 [Short] Why is calcium hydroxide added to seawater in the Dow process?

Magnesium hydroxide is much less soluble than calcium hydroxide. Adding $Ca(OH)_2$ therefore precipitates the $Mg^{2+}$ ions as insoluble $Mg(OH)_2$, which can be filtered off from the seawater.

### Q.54 [Long] Describe the production of quicklime from limestone.

Quicklime (calcium oxide, CaO) is produced by heating limestone ($CaCO_3$), a naturally abundant sedimentary rock, in a high-temperature furnace called a **lime kiln** at about 900–1000 °C. The process is called **calcination** or thermal decomposition.

$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)} \text{ (heat)}$$

In modern kilns the ingredients are heated by circulating hot gases. The reaction is reversible in a closed container, but $CO_2$ is removed continuously. This shifts the equilibrium to the right, giving faster production of a large amount of product. Quicklime is used in construction (cement, mortar), agriculture (soil treatment), metallurgy and chemical industries.

### Q.55 [Short] Why is carbon dioxide removed continuously from a lime kiln?

The decomposition of $CaCO_3$ is reversible in a closed container. Removing $CO_2$ shifts the equilibrium to the right (Le Chatelier's principle), so the reaction goes faster and gives a large amount of CaO.

### Q.56 [Short] What is calcination?

Calcination is the heating of a substance such as limestone to a very high temperature (900–1000 °C) in a kiln so that it decomposes thermally.

## 17.8 USES OF GROUP 2 COMPOUNDS

### Q.57 [Long] Describe the role of Group 2 compounds in industry and everyday life.

- **Construction:** $CaCO_3$ (limestone, marble, chalk) is a building material, used in cement and concrete and as a filler in paper and plastics. CaO (quicklime) and $Ca(OH)_2$ (slaked lime) are used in cement, mortar and plaster. Quicklime removes impurities in steel production. Calcium sulphate ($CaSO_4 \cdot 2H_2O$, gypsum) is used for plasterboard and plaster of Paris ($CaSO_4 \cdot ½H_2O$).
- **Environment:** CaO and $CaCO_3$ are used in flue gas desulphurization (wet scrubbing) to remove acidic $SO_2$ from power plant emissions and prevent acid rain. MgO is a refractory material (furnace lining) due to its high melting point.
- **Metals and their compounds:** Magnesium is used in lightweight alloys (aerospace, automotive), fireworks and flares. Strontium compounds give the red colour in fireworks and flares. Beryllium is used in alloys to increase strength (springs, aerospace) and in X-ray tubes, as it is transparent to X-rays.

### Q.58 [Long] Describe the uses of Group 2 compounds in medicine.

- $Mg(OH)_2$ ("Milk of Magnesia") neutralises excess stomach acid and is a mild laxative; its low solubility makes it safe.
- $CaCO_3$ is used in indigestion tablets as an antacid. Calcium compounds (carbonate, citrate) are dietary supplements for strong bones and teeth and prevent osteoporosis.
- Magnesium compounds (sulphate, citrate) are important for muscle and nerve function and treat magnesium deficiency.
- Epsom salts ($MgSO_4 \cdot 7H_2O$) are used as a laxative and in baths.
- $BaSO_4$ is used in "barium meals" for X-ray imaging of the digestive system. It is opaque to X-rays and safe because it is highly insoluble and passes through the body without being absorbed.

**Q.59** [Short] **Describe the uses of Group 2 compounds in agriculture.**

- $CaCO_3$ and $Ca(OH)_2$ are added to acidic soils to increase the pH (neutralise acidity), making the soil more suitable for crops.
- Calcium and magnesium are essential plant nutrients: calcium for cell wall development; magnesium is the central atom of chlorophyll, vital for photosynthesis.
- Calcium nitrate and magnesium sulphate (Epsom salts) can be used as fertilizers.

**Q.60** [Short] **Barium compounds are poisonous, yet a "barium meal" is safe. Why?**

$BaSO_4$ is highly insoluble. It passes through the body without being absorbed, so no barium ions enter the blood.

**Q.61** [Short] **What is flue gas desulphurization?**

It is the removal of acidic sulphur dioxide ($SO_2$) from power plant emissions by wet scrubbing with CaO or $CaCO_3$. It prevents acid rain.

**Q.62** [Short] **Differentiate between quicklime, slaked lime and limestone.**

| Name | Formula | Use |
|---|---|---|
| Limestone | $CaCO_3$ | Building material; cement; raw material for quicklime |
| Quicklime | CaO | Cement, mortar; removes impurities in steel making |
| Slaked lime | $Ca(OH)_2$ | Mortar, plaster; neutralising acidic soils |

## Quick Check 17.4

**Q.63** [Short] **a) Write the balanced equation for the electrolysis step in the Dow Process.**

$$Mg^{2+}_{(l)} + 2Cl^{-}_{(l)} \rightarrow Mg_{(l)} + Cl_{2(g)}$$

**Q.64** [Short] **b) Write the balanced equation for producing quicklime (CaO) from limestone.**

$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}\ (heat)$$

**Q.65** [Short] **c) State one agricultural use of $CaCO_3$ and the chemical principle involved.**

It is added to acidic soils to raise the pH. The principle is neutralisation: $CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$.